# Short | Box-Cox gamma sweep | Val ANN10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 12/18 |
| **1 ANN10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 2/18 |
| **1 ANN10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **1 ANN10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **1 ANN10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **1 ANN10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val ANN10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 10/18 |
| **1 ANN10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 4/18 |
| **1 ANN10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 3/18 |
| **1 ANN10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **1 ANN10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **1 ANN10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2 P-only DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 4/18 |
| **2 P-only DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 4/18 |
| **2 P-only DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 8/18 |
| **2 P-only DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 3/18 |
| **2 P-only DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **2 P-only DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2 P-only DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 3/18 |
| **2 P-only DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 3/18 |
| **2 P-only DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 9/18 |
| **2 P-only DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 5/18 |
| **2 P-only DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **2 P-only DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val ANN10 + P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **3 ANN10 + P-only DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 6/18 |
| **3 ANN10 + P-only DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 2/18 |
| **3 ANN10 + P-only DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 9/18 |
| **3 ANN10 + P-only DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 5/18 |
| **3 ANN10 + P-only DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **3 ANN10 + P-only DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val ANN10 + P-only DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **3 ANN10 + P-only DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **3 ANN10 + P-only DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 2/18 |
| **3 ANN10 + P-only DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 8/18 |
| **3 ANN10 + P-only DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **3 ANN10 + P-only DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **3 ANN10 + P-only DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **4 T/P DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 3/18 |
| **4 T/P DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **4 T/P DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **4 T/P DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **4 T/P DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **4 T/P DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **4 T/P DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 6/18 |
| **4 T/P DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 6/18 |
| **4 T/P DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 14/18 |
| **4 T/P DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **4 T/P DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **4 T/P DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val ANN10 + T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **5 ANN10 + T/P DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 5/18 |
| **5 ANN10 + T/P DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **5 ANN10 + T/P DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 6/18 |
| **5 ANN10 + T/P DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **5 ANN10 + T/P DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **5 ANN10 + T/P DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val ANN10 + T/P DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **5 ANN10 + T/P DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 5/18 |
| **5 ANN10 + T/P DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **5 ANN10 + T/P DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 8/18 |
| **5 ANN10 + T/P DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **5 ANN10 + T/P DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **5 ANN10 + T/P DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val T/P/$\phi$ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **6 T/P/$\phi$ DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 6/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 2/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **6 T/P/$\phi$ DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val T/P/$\phi$ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **6 T/P/$\phi$ DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 11/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 8/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **6 T/P/$\phi$ DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **6 T/P/$\phi$ DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Short | Box-Cox gamma sweep | Val ANN10 + T/P/$\phi$ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 10/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 4/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

# Long | Box-Cox gamma sweep | Val ANN10 + T/P/$\phi$ DeepONet10: common ICs []

Raw | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

| Model / same-state mean | gamma | T | H2 | H | O | O2 | OH | H2O | HO2 | H2O2 | N2 | 10 ms status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.05 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 7/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 13/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 0.5 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 0/18 |

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.